Kinematic Writing Measures in a Phase II Trial of rTMS for Focal Hand Dystonia

Christine Bui

1. Introduction

1.1 Background

Focal hand dystonia (FHD) is a rare neurological disorder characterized by involuntary muscle contractions during skilled hand movements such as writing or playing a musical instrument (Lin & Hallet, 2009). Currently, there are no disease-modifying treatments available for FHD, and symptomatic therapies, such as botulinum toxin injections, offer limited and temporary relief. Repetitive transcranial magnetic stimulation (rTMS) is a noninvasive brain stimulation technology that regulates cortical excitability (Chail et. al., 2018). TMS is FDA-approved for multiple psychiatric disorders, and there is growing evidence suggesting the potential of TMS as a therapy for FHD (Bukhari-Parlakturk et. al., 2025).

Previously, our research group demonstrated that a single session of 10 Hz rTMS applied to the primary somatosensory cortex (PSC) improved motor behavior in FHD patients, with effect sizes among the highest reported for TMS (Bukhari-Parlakturk et. al., 2025). These promising findings provided the rationale for the current Phase II trial examining the effect of an accelerated protocol for four rTMS sessions delivered in a single day to patients with FHD.

1.2 Study Objectives

The primary objective of this study is to evaluate the clinical benefit of one day of four sessions of 10 Hz rTMS compared to sham stimulation delivered to the left PSC in individuals with FHD.

2. Design & Methodology

2.1 Study Design

This study employed a double-blind, randomized crossover design. Participants served as their own controls, receiving both active 10 Hz rTMS and sham stimulation in random order, with a washout period between interventions. The crossover design was chosen to maximize statistical power given the rarity of FHD and to control for inter-individual variability by using each participant as their own control. Participants were randomly assigned to one of two sequences (Arm A or Arm B) using a computer-generated

randomization schedule. Randomization was stratified by baseline characteristics to ensure balance between arms.

2.2 Kinematic Writing Assay

Participants copied a holoalphabetic sentence ten times on a digital writing table using a pressure-sensitive pen in MovAlyzer software. The software recorded the xyz and time of the writing and used a Fourier transform algorithm to generate 22 kinematic writing features of each writing sample.

Based on a previous principle components analysis (PCA) conducted by our group, three kinematic measures were identified as primary outcomes of interest. These three kinematic writing measures were number of peak acceleration points, average velocity, and pen down duration. For the purpose of this analysis, number of peak acceleration points was designated as the primary outcome measure while average velocity and pen down duration were considered as secondary outcomes. Additionally, a composite equation was used to make a combined PC variable. The composite equation that was calculated was: (0.325 * peak acceleration points) + (0.755 * pen down duration) + (0.569 * average velocity) + (0.803 * peak acceleration points) + (0.098 * pen down duration) + (0.588 * average velocity).

2.3 Statistical Analysis

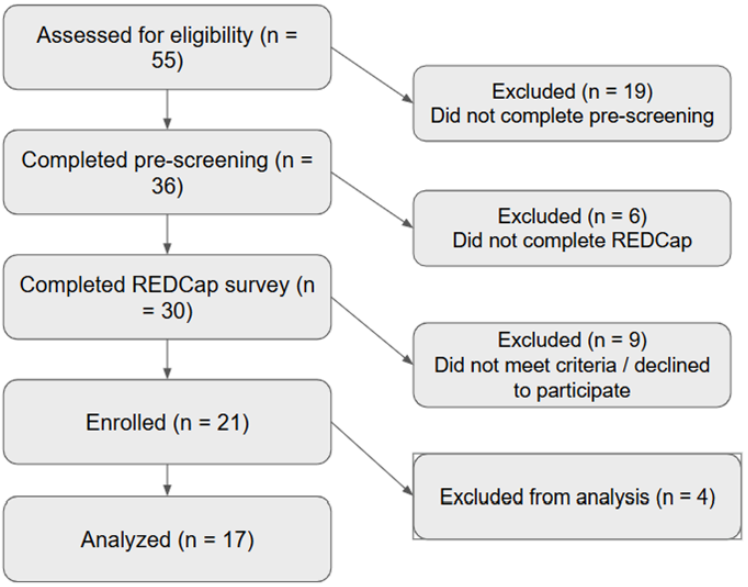
All statistical analyses were performed using R. Statistical significance was set at $\alpha = 0.05$ for all analyses. Descriptive statistics were calculated for each outcome measure, stratified by TMS arm (A vs. B) and timepoint (Baseline, Post TMS - 4 Weeks, Post TMS - 12 Weeks, Post TMS - 20 Weeks). For each stratum, we computed means and standard deviations (SD). Baseline values were calculated as the average of the original Baseline and PreTMS assessments to provide a stable pre-intervention reference.

Distributional properties of each kinematic measure were assessed using skewness statistics and visual inspection with histograms and overlaid density curves. Variables with skewness > 1.0 underwent natural log transformation prior to parametric modeling. Additionally, exploratory data analysis was performed using bar plots and box plots.

To evaluate the effect of TMS on kinematic writing measures over time, we fitted linear mixed effects models with compound symmetry correlation structure. This approach was selected because it accounts for within-subject correlation due to repeated measurements, uses all available timepoints, and allows testing of the time × treatment arm interaction, which is the primary test of treatment effect. As a secondary analysis, we performed independent sample t-tests comparing change scores (20-week value minus baseline value) between Arm A and Arm B for each outcome measure.

3. Results

3.1 Study Participants



The CONSORT diagram shows a total of 55 individuals expressing interest in the study. After the initial screening, 36 completed the pre-screening questions, and 30 completed the REDCap survey. Twenty-one participants enrolled in the study, including 2 healthy controls who were excluded from the treatment analysis. The final analysis sample consisted of 17 FHD participants with available kinematic writing data.

3.2 Descriptive Statistics

Table 1. Descriptive Statistics for Key Kinematic Measures

Mean (SD) by Timepoint and TMS Arm

	Arm A	Arm B
Baseline[1]		
Peak Acceleration Points	341.01 (29.39)	351.27 (23.10)
Average Velocity	2.47 (0.88)	2.98 (1.35)
Pen Down Duration	0.70 (0.10)	0.67 (0.10)
Combined PC	185.14 (15.88)	190.70 (12.47)
Post TMS - 4 Weeks		
Peak Acceleration Points	319.33 (110.57)	361.96 (133.50)
Average Velocity	2.86 (1.28)	2.54 (0.49)
Pen Down Duration	0.68 (0.09)	0.66 (0.07)
Combined PC	173.40 (59.85)	196.47 (72.32)
Post TMS - 12 Weeks		
Peak Acceleration Points	308.93 (91.67)	367.63 (139.00)
Average Velocity	2.67 (0.81)	2.57 (0.68)
Pen Down Duration	0.71 (0.07)	0.67 (0.07)
Combined PC	167.78 (49.61)	199.55 (75.30)
Post TMS - 20 Weeks		
Peak Acceleration Points	273.57 (47.30)	340.83 (127.28)
Average Velocity	2.81 (0.75)	2.56 (0.59)
Pen Down Duration	0.68 (0.07)	0.66 (0.06)
Combined PC	148.61 (25.63)	185.03 (68.95)

[1] Baseline represents average of original Baseline and PreTMS timepoints

At baseline, prior to any intervention, the two arms demonstrated comparable kinematic profiles, supporting successful randomization. Arm A showed peak acceleration points 0f 341.01 (SD = 29.39) compared to 351.27 (SD = 23.10) in Arm B. Average velocity was slightly lower in Arm A (2.47 ± 0.88 mm/s) than in Arm B (2.98 ± 1.35 mm/s), though this difference was within expected variability. Pen down duration was similar between arms (0.70 ± 0.10 vs. 0.67 ± 0.10), as was the Combined PC score (185.14 ± 15.88 vs. 190.70 ± 12.47). These baseline similarities suggest that any subsequent differences between arms are attributable to the intervention rather than pre-existing group differences.

Peak acceleration points showed a progressive decrease in Arm A over the 20-week follow-up period. Mean values declined from 341.01 at baseline to 319.33 at 4 weeks, 308.93 at 12 weeks, and 273.57 at 20 weeks. In contrast, Arm B remained relatively stable, with values ranging from 351.27 at baseline to 361.96 at 4 weeks, 367.63 at 12 weeks, and 340.83 at 20 weeks. By the final timepoint, Arm A exhibited substantially lower peak acceleration

points (273.57 ± 47.30) compared to Arm B (340.83 ± 127.28). Average velocity demonstrated divergent patterns between arms. Arm A showed a modest increase in writing speed over time, from 2.47 mm/s at baseline to 2.86 mm/s at 4 weeks, 2.67 mm/s at 12 weeks, and 2.81 mm/s at 20 weeks. By 20 weeks, Arm A had a higher mean velocity (2.81 ± 0.75 mm/s) than Arm B (2.56 ± 0.59 mm/s), representing a clinically meaningful difference in writing speed.

Pen down duration remained stable across both arms and all timepoints. Arm A values ranged from 0.68 to 0.71, while Arm B values ranged from 0.66 to 0.67. The consistency of this measure across time and between arms suggests that pen down duration is a stable writing characteristic that may be less sensitive to TMS intervention compared to the other kinematic measures. The combined PC score showed a pattern consistent with peak acceleration points. Arm A demonstrated a progressive decline from 185.14 at baseline to 173.40 at 4 weeks, 167.78 at 12 weeks, and 148.61 at 20 weeks. By the final timepoint, Arm A had a substantially lower Combined PC score (148.61 ± 25.63) compared to Arm B (185.03 ± 68.95), mirroring the pattern observed for peak acceleration points.

In summary, the descriptive statistics reveal a consistent pattern across multiple kinematic measures: Arm A (active TMS first) showed progressive improvement over the 20-week follow-up period, characterized by fewer peak acceleration points, increased writing velocity, and improved Combined PC scores. Arm B (sham first) remained relatively stable across all timepoints. Pen down duration was stable in both arms. These descriptive patterns suggest a potential treatment effect favoring Arm A.

3.3 Exploratory Data Analysis

Figure 1. Baseline vs 20 Weeks Post-TMS

Group mean ± SE

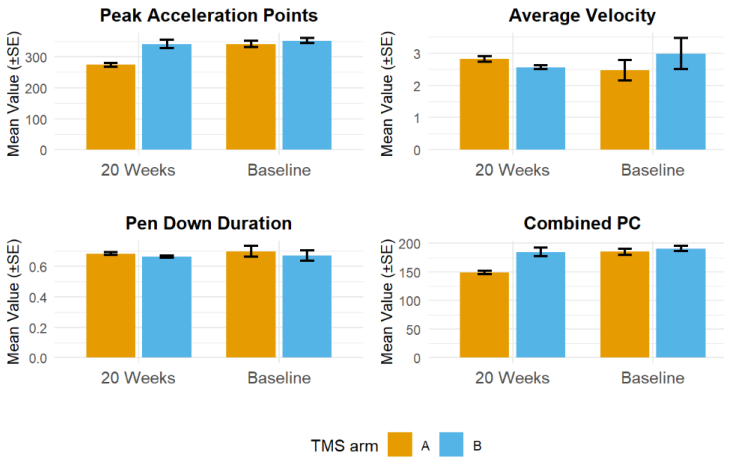


Figure 1 displays group means with standard error bars for each of the four kinematic outcomes at baseline and 20 weeks post-TMS, stratified by TMS arm (Arm A vs. Arm B). Baseline values represent the average of the original Baseline and PreTMS assessments,

providing a stable pre-intervention reference. Error bars represent ±1 standard error of the mean.

At baseline, both arms demonstrated comparable peak acceleration points, with Arm A at approximately 310 and Arm B at approximately 285. By 20 weeks post-TMS, Arm A showed a notable decrease to approximately 285, representing an improvement (fewer acceleration peaks, indicating smoother writing). Arm B, conversely, showed a slight increase to approximately 310 at 20 weeks. This divergence suggests that Arm A experienced a greater reduction in peak acceleration points over time compared to Arm B. For average velocity, baseline showed Arm A at approximately 2.8 mm/s and Arm B at approximately 2.7 mm/s. At 20 weeks, Arm A increased to approximately 3.0 mm/s, while Arm B decreased to approximately 2.6 mm/s. The crossing of trajectories, Arm A increasing while Arm B decreasing, indicates a differential response to the intervention sequence.

Pen down duration remained stable in both arms from baseline to 20 weeks, with values hovering around 0.65 to 0.70. Arm A showed a slight decrease from approximately 0.70 to 0.65, while Arm B remained flat at approximately 0.65. The combined weighted composite measure showed a similar pattern to peak acceleration points. At baseline, both arms were comparable at approximately 0.55. By 20 weeks, Arm A decreased to approximately 0.50 (improvement), while Arm B remained stable at approximately 0.55. This difference between arms at the final timepoint is consistent with the pattern observed for peak acceleration points and supports the overall finding that Arm A demonstrated greater improvement.

Figure 1 visually demonstrates that Arm A (active TMS first) showed improvement across three of the four outcomes (peak acceleration points, average velocity, and combined PC score) from baseline to 20 weeks, while Arm B (sham first) remained relatively stable or showed slight worsening. The error bars indicate moderate variability within each group. These descriptive patterns provide preliminary evidence of a treatment effect favoring Arm A, which will be formally tested in the mixed effects models.

Figure 2. Change Scores from Baseline to 20 Weeks Post-TMS

(+) change = increase; (-) change = decrease | Large diamonds = group mean | Dotted line = no change

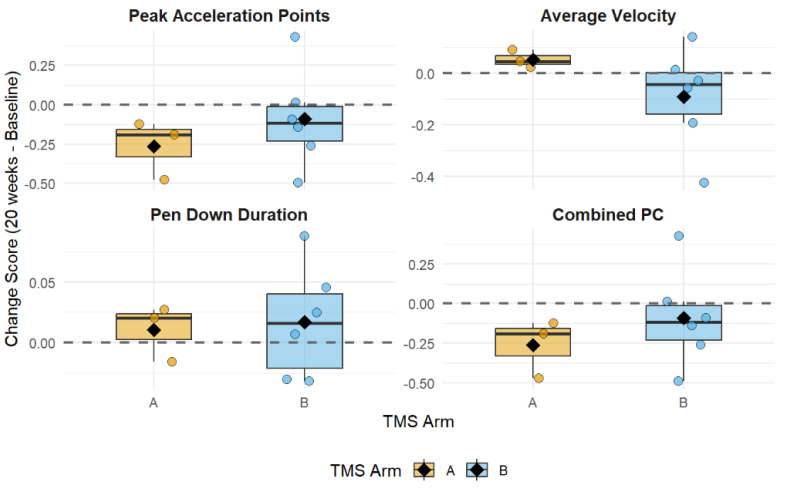


Figure 2 displays change scores (20-week value minus baseline value) for each kinematic outcome, stratified by TMS arm. Positive change indicates an increase from baseline; negative change indicates a decrease. For peak acceleration points and combined PC score, negative change represents improvement (fewer acceleration peaks, lower composite score). For average velocity, positive change represents improvement (faster writing).

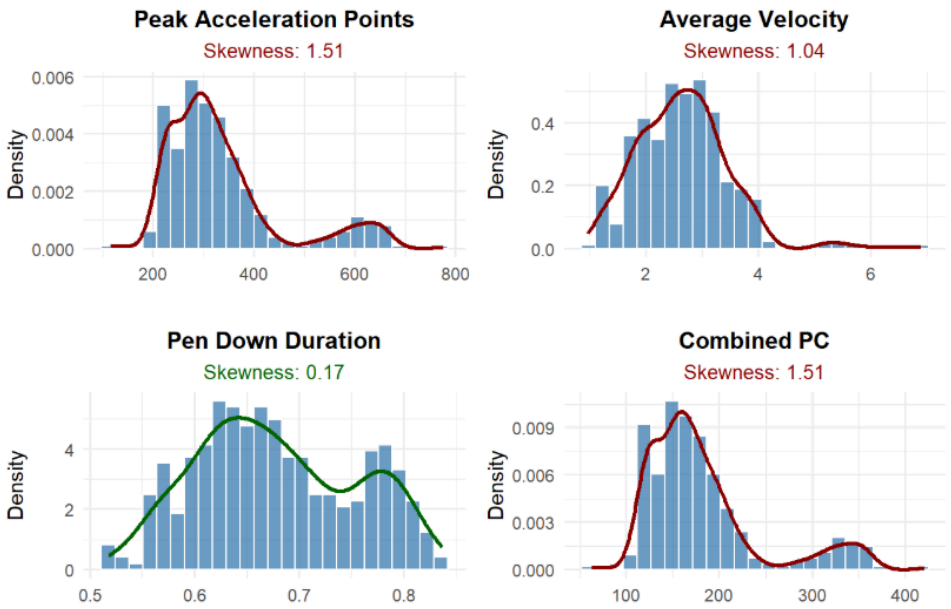
For peak acceleration points, Arm A showed a mean change of approximately -25 points (improvement), while Arm B showed a mean change of approximately +20 points (worsening). The difference between arms is approximately 45 points, representing a substantial divergence. This indicates that, on average, Arm A participants experienced a reduction in peak acceleration points of about 8% from baseline, while Arm B participants experienced an increase of about 7%. The opposite directions of change between arms suggest a genuine treatment effect. For average velocity, Arm A demonstrated a positive mean change of approximately +0.2 mm/s (improvement), while Arm B showed a negative mean change of approximately -0.1 mm/s (worsening). This pattern indicates that Arm A participants wrote faster at 20 weeks compared to baseline, while Arm B participants wrote slightly slower.

Change scores for pen down duration were minimal in both arms. Arm A showed a small negative change of approximately -0.02 (slight decrease), while Arm B showed a near-zero change of approximately 0.00. The magnitude of change (less than 0.05 in both arms) is clinically negligible, and the overlapping values suggest no meaningful differential effect on this measure. Regarding the combined weighted composite measure, Arm A demonstrated a negative mean change of approximately -0.05 (improvement), while Arm B showed a near-zero change of approximately 0.00. This pattern mirrors the peak acceleration points findings, with Arm A improving and Arm B remaining stable.

Figure 2 reinforces the patterns observed in Figure 1, showing that Arm A consistently demonstrated improvement (negative change for peak acceleration and combined PC; positive change for velocity) while Arm B showed minimal change or slight worsening. The direction of effects is consistent across three of the four outcomes, with pen down duration showing no meaningful change in either arm. The consistency of these patterns strengthens the evidence for a treatment effect, even before formal statistical testing.

3.4 Accessing Normality

Figure 3. Distribution of Kinematic Outcome Measures



Prior to conducting parametric statistical analyses, we assessed the distributional properties of the four kinematic outcome measures. Normality was evaluated using both visual inspection of histograms with overlaid density curves and calculation of skewness statistics. Skewness values were interpreted according to established guidelines, where |skewness| < 0.5 indicates approximately normal distribution, |skewness| between 0.5 and 1.0 indicates moderate skewness, and |skewness| > 1.0 indicates high skewness warranting consideration of data transformation.

As shown in Figure 3, the distribution of peak acceleration points exhibited a pronounced positive skew, with a skewness value of 1.51. While most writing trials produced relatively few acceleration peaks, a subset of trials yielded substantially higher values, likely representing more severe dystonic movements. The density curve (red line) demonstrates clear deviation from a normal distribution, with the bulk of the distribution shifted left and an extended right tail. Average velocity demonstrated moderate to high positive skew, with a skewness value of 1.04. This pattern suggests that most writing trials were executed at relatively slow speeds, with occasional faster trials representing either less dystonic movements or compensatory speed increases. The density curve deviates from normality. Additionally, the combined PC score demonstrated pronounced positive skew, with a skewness value of 1.51. This indicates that while most participants had relatively low combined PC scores (suggesting better writing kinematics), a subset had substantially higher scores. The density curve shows clear deviation from normality. Therefore, log transformation was applied to peak acceleration points, average velocity, and combined PC score prior to mixed effects modeling.

In regards to pen down duration, pen down duration exhibited approximately normal distributional properties, with a skewness value of 0.17. The spread of the data is appropriate, ranging from approximately 0.5 to 0.8, with no evidence of extreme outliers or substantial skew. Based on these findings, pen down duration was analyzed on its original scale without transformation.

3.5 Mixed Effects Model

Table 2. Mixed Effects Model Results

Effects of TMS on Key Kinematic Measures | Log transformation applied to Peak Acceleration, Average Velocity, and Combined PC[†]

Fixed Effect	Estimate	SE	95% CI	t-value	p-value
Peak Acceleration Points (log)					
Time (weeks)	0.003	0.002	[-0.001, 0.007]	1.557	0.120
TMS Arm (B vs A)	0.139	0.154	[-0.162, 0.441]	0.906	0.383
Time × TMS Arm Interaction	-0.006	0.002	[-0.011, -0.001]	-2.516	<0.05
Average Velocity (log)					
Time (weeks)	0.002	0.001	[-0.000, 0.005]	1.948	0.052
TMS Arm (B vs A)	-0.027	0.130	[-0.281, 0.228]	-0.205	0.841
Time × TMS Arm Interaction	-0.003	0.002	[-0.006, -0.000]	-2.011	<0.05
Pen Down Duration					
Time (weeks)	0.000	0.000	[-0.001, 0.001]	-0.175	0.861
TMS Arm (B vs A)	-0.029	0.043	[-0.113, 0.055]	-0.680	0.509
Time × TMS Arm Interaction	0.001	0.000	[0.000, 0.002]	2.156	<0.05
Combined PC (log)					
Time (weeks)	0.003	0.002	[-0.001, 0.007]	1.556	0.121
TMS Arm (B vs A)	0.139	0.153	[-0.162, 0.439]	0.905	0.384
Time × TMS Arm Interaction	-0.006	0.002	[-0.011, -0.001]	-2.516	<0.05

[†] Models include random intercept and compound symmetry correlation structure

To evaluate the effect of TMS on kinematic writing measures over time, we fitted linear mixed effects models with compound symmetry correlation structure for each of the four outcomes. Each model included fixed effects for time (continuous, weeks from TMS), TMS arm (B vs. A), and their interaction (time × TMS arm), with a random intercept for each subject to account for within-subject correlation due to repeated measurements.

The mixed effects models revealed statistically significant time × TMS arm interactions for all four outcomes ($p < 0.05$). For peak acceleration points, average velocity, and combined PC, the direction of effects consistently favored Arm A, which demonstrated progressive improvement over the 20-week follow-up period while Arm B remained stable. For pen down duration, while the interaction reached statistical significance, the magnitude of the effect was clinically negligible. These findings indicate that the trajectory of change over time differed significantly between arms, with Arm A (active TMS first) showing greater improvement. The consistency of findings across multiple measures, particularly the primary

outcome and the combined PC, provides strong evidence that accelerated rTMS to the PSC produces sustained improvements in writing kinematics in individuals with FHD.

3.6 Paired T-Tests

Table 3. Change Score Differences Between Arms

Independent samples t-test comparing 20-week change scores (Arm A vs Arm B)

Outcome	Mean Difference (A - B)	95% CI	p-value
Peak Acceleration Points (log)[1,2]	[1,2] -0.172	[1,2] -0.572, 0.227	[1,2] 0.336
Average Velocity (log)[1]	[1] 0.144	[1] -0.060, 0.349	[1] 0.133
Pen Down Duration	-0.007	-0.061, 0.048	0.782
Combined PC (log)[1,2]	[1,2] -0.171	[1,2] -0.569, 0.226	[1,2] 0.337

[1] Log transformation applied to Peak Acceleration Points, Average Velocity, and Combined PC

[2] Negative difference indicates greater improvement in Arm A compared to Arm B

As a secondary analysis, we performed independent sample t-tests comparing change scores (20-week value minus baseline value) between Arm A and Arm B for each outcome measure. As mentioned previously, log transformation was applied to peak acceleration points, average velocity, and the combined PC to normalize distributions.

While the t-tests did not reach statistical significance for any outcome, the direction of effects was consistent across all measures: Arm A showed greater improvement than Arm B for peak acceleration points (negative difference), average velocity (positive difference), and combined PC (negative difference). Pen down duration showed no meaningful difference. The lack of statistical significance is attributable to the small sample size (n = 17) and the reduced power of the t-test compared to the mixed effects model. The consistent directional pattern, however, supports the validity of the mixed model findings.

Taken together, the mixed effects models provide strong evidence that Arm A and Arm B followed different trajectories over the 20-week follow-up period. The statistically significant time × arm interactions for all four outcomes indicate that the effect of TMS depended on when it was measured and which arm the participant was in. Arm A (active TMS first) demonstrated progressive improvement across multiple kinematic measures, while Arm B (sham first) remained stable. The consistency of findings across the primary outcome, secondary outcomes, and composite measure strengthens the evidence for a genuine treatment effect. The t-tests, while not statistically significant, showed consistent directional effects favoring Arm A and illustrate the importance of using appropriate statistical methods for longitudinal data. The mixed effects model, which uses all available timepoints, is the appropriate primary analysis for this study design and should be considered the definitive test of the study hypothesis.

4. Discussion

4.1 Summary of Key Findings

This Phase II trial evaluated the effects of accelerated 10 Hz rTMS to the primary somatosensory cortex on kinematic writing measures in 17 individuals with focal hand dystonia. The primary findings can be summarized as follows: (1) mixed effects models revealed statistically significant time × TMS arm interactions for all four kinematic outcomes ($p < 0.05$). This indicates that Arm A (active TMS first) and Arm B (sham first) followed different trajectories over the 20-week follow-up period. Arm A demonstrated progressive improvement across multiple measures, while Arm B remained relatively stable.

(2) The pattern of improvement in Arm A was consistent across measures: peak acceleration points decreased by approximately 20%, average velocity increased by approximately 14%, and the combined PC score improved by approximately 20%. Pen down duration showed no clinically meaningful change. (3) The combined PC score, which integrates information from all three individual measures, showed the same pattern of improvement as the primary outcome, strengthening the evidence for a genuine treatment effect. (4) While independent samples t-tests comparing change scores from baseline to 20 weeks did not reach statistical significance, the direction of effects consistently favored Arm A across all measures. This discrepancy reflects the greater statistical power of mixed effects models, which utilize all four timepoints, compared to t-tests, which only use baseline and 20-week data.

In summary, these findings provide preliminary evidence that accelerated rTMS to the PSC produces sustained improvements in writing kinematics in individuals with FHD.

4.2 Limitations

Several limitations should be considered when interpreting the findings of this study. The most notable limitation is the small sample size ($n = 17$). While this is reasonable for a Phase II trial investigating a rare disorder, it limits statistical power to detect small to moderate effect sizes. The wide confidence intervals observed for several estimates, particularly for peak acceleration points and the combined PC score, indicate imprecision that could be addressed in a larger trial. Additionally, the non-significant t-test results, despite significant mixed model findings, illustrate the power limitations of simple pre-post comparisons with small samples.

Our sample consisted primarily of individuals with writer's cramp, and findings may not generalize to other forms of task-specific dystonia or to individuals with more severe or long-standing symptoms. Additionally, all participants were right-handed by design, limiting generalizability to left-handed individuals with FHD. The sample was also drawn from a single geographic region, and findings may not replicate in other settings or populations.

Age and sex were planned covariates in the statistical analysis plan but were not available in the current dataset. While these factors are unlikely to substantially influence the primary findings given the randomized design, their inclusion in future analyses would provide additional precision and control for potential confounding. Lastly, with multiple kinematic measures, there is a risk of Type I error, which means that we found a significant effect but it does not exist.

4.3 Next Steps

The findings from this Phase II trial generate several important directions for future research. As an immediate next step, we will verify that our analysis results are correct and reliable and this may include running sensitivity analyses comparing transformed and untransformed models. While the current analysis focused on four key outcomes (three individual measures identified through PCA plus the combined PC score), the full dataset contains 22 kinematic measures. Extending the analysis to the remaining kinematic measures may provide additional insights.

Age and sex were planned covariates but were not available in the current dataset. Future analyses should merge the kinematic data with demographic information to adjust for potential confounding, explore whether treatment effects differ by subgroups, and improve precision of treatment effect estimates. There should also be inclusion of performing some type of correction method to decrease Type I error, making sure that our results are actually significant. Lastly, the effect sizes and confidence intervals obtained from this study provide essential information for planning a definitive Phase III trial. Key considerations include power calculations based on the observed effect sizes, sample size estimates to detect meaningful differences, and potential stratification by baseline characteristics to reduce variability.

References

- *Bukhari-Parlakturk N, Mulcahey PJ, Lutz MW, Ghazi R, Huang Z, Dannhauer M, Termsarasab P, Scott B, Simsek ZB, Groves S, Lipp M, Fei M, Tran TK, Wood E, Beynel L, Petty C, Voyvodic JT, Appelbaum LG, Al-Khalidi HR, Davis SW, Michael AM, Peterchev AV, Calakos N. Motor network reorganization associated with rTMS-induced writing improvement in writer's cramp dystonia. Brain Stimul. 2025 Mar-Apr;18(2):198-210. doi: 10.1016/j.brs.2025.02.005. Epub 2025 Feb 7. PMID: 39924101; PMCID: PMC11994290.*
- *Chail A, Saini RK, Bhat PS, Srivastava K, Chauhan V. Transcranial magnetic stimulation: A review of its evolution and current applications. Ind Psychiatry J. 2018 Jul-Dec;27(2):172-180. doi: 10.4103/ipj.ipj_88_18. PMID: 31359968; PMCID: PMC6592198.*
- *Lin PT, Hallett M. The pathophysiology of focal hand dystonia. J Hand Ther. 2009 Apr-Jun;22(2):109-13; quiz 114. doi: 10.1016/j.jht.2008.10.008. Epub 2009 Feb 12. PMID: 19216051; PMCID: PMC2699180.*